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# Forecasting plant community impacts of climate variability and change: when do competitive interactions matter?

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## Summary

**1**. A change in a climate variable may alter a species' abundance not only through a direct effect on that species' vital rates, but also through 'indirect' effects mediated by species interactions. While recent work has highlighted cases in which indirect effects overwhelm the direct effects of climate, we lack robust generalizations to predict the strength of indirect effects.

**2**. For communities dominated by non-trophic interactions, we propose that the potential for indirect effects of climate change declines with the strength of stabilizing niche differences.

**3**. We tested this hypothesis by analysing an empirically parameterized four species population model. We quantified negative frequency dependence in population growth rates as a measure of stabilizing niche differences and projected the sensitivity of each species to direct and indirect effects of climate perturbations.

**4**. Consistent with our hypothesis, species' sensitivities to indirect effects decreased rapidly with increasing stabilization by niche differences.

**5**. *Synthesis*. Information about niche differences can identify species sensitive to indirect effects of climate change and determine when multispecies forecasting approaches are necessary. However, practical application of this generalization will require methods to predict niche differences from easily collected data.

**Key-words:** climate change, community ecology, competition, demography, integral projection model, niche, plant population and community dynamics, sagebrush steppe

## Introduction

Understanding how climatic variation affects population and community dynamics, one of the classic problems in ecology (Andrewartha & Birch 1954; Nicholson 1954) has taken on new urgency given the need to forecast the impacts of anthropogenic climate change (Clark *et al.* 2001; Carpenter 2002). One of the greatest sources of uncertainty in predicting how communities will respond to climate change is the role of species interactions. A change in a climate variable could alter a species' abundance not only through a 'direct' effect on that species' vital rates, but also through 'indirect' effects mediated by species interactions. These indirect effects of climate occur when an altered climate variable influences the abundance of other species in the community (through direct effects on those species' vital rates) or influences their per capita interaction effects. Recent work has shown that such indirect effects may be strong, potentially reversing the direct effects of climate (Brooker 2006; Suttle, Thomsen & Power 2007; Tylianakis *et al.* 2008; Gilman *et al.* 2010). In communities where indirect effects are important, ecological forecasts must be based on explicit modelling of species interactions. Unfortunately, we currently lack robust generalizations that would predict when indirect effects of climate change are likely to be strong versus weak (Gilman *et al.* 2010).

While the responses of individual species to particular climate variables may be idiosyncratic, we believe that robust generalizations are already available in the theoretical literature if we focus on a more fundamental question: What determines the strength of species interactions? Research on interaction strengths and their consequences for community dynamics has a long tradition in community ecology (e.g. Wootton 1994). Findings from this research could help us to determine when we can ignore species interactions and when we need to model them explicitly to accurately predict the community impacts of climate change. For example,

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Murdoch *et al.* (2002) showed that single-species models adequately describe the population dynamics of generalist, but not specialist, consumer species in many species food webs. The implication is that forecasting models for specialist consumers must include prey dynamics and the resulting indirect effects of climate drivers, whereas for generalist species models that only include direct effects of climate and treat species interactions implicitly might be adequate. Other recent studies have shown that much of the variation in the observed strength of pairwise, consumer–resource interactions is explained by biomass and body size (Berlow *et al.* 2009; Wood *et al.* 2010). This work suggests that large-bodied organisms are more likely to exert strong indirect effects under climate change.

Our goal was to apply coexistence theory to identify similar, general predictions for the indirect effects of climate change resulting from non-trophic interactions, such as competition among plant species. Niche differences promote diversity by weakening interspecific interactions relative to intraspecific interactions (Chesson 2000). In the extreme case, two species occupying totally distinct niches will have independent dynamics. Therefore, the stronger the niche differentiation between a focal species and the rest of the plant community, the weaker the potential indirect effects of climate. Conversely, weak niche differentiation (high niche overlap) would lead to strong indirect effects. Our analysis of a two-species annual plant model and general size-structured multispecies models (Table 1 and Appendix S1 in Supporting Information) suggests that this intuitive logic is sound. However, an empirical test of the hypothesis presents two daunting challenges: We need estimates of the indirect effects of climate variables and a quantitative description of niche differences for multiple species in a natural plant community. Empirical estimates of either quantity are exceedingly rare, and the combination of both types of information for one community is, to our knowledge, unprecedented.

We are in a position to conduct an empirical test of the relationship between niche differences and the indirect effects of climate thanks to long-term, mapped data collected in the first half of the 20th century in a sagebrush steppe plant community. These data allow us to model the influence of observed climatic variation on species' intrinsic demographic rates and also to characterize species interactions and niche differences. We begin by fitting a multispecies population model in which interannual variation in vital rates is explained by climate variables. We then perturb the observed climate variables to estimate the sensitivity of each species' abundance to direct and indirect effects. Next, we estimate the degree of niche differentiation among species based on the simulations of negative frequency dependence in per capita growth rates (Adler, HilleRisLambers & Levine 2007). Finally, we show that this empirical test supports our general theoretical prediction of a strong inverse relationship between the strength of stabilizing niche differences and species' sensitivity to the indirect effects of climate changes.

## Materials and methods

Our analysis proceeds in four steps: (i) extract demographic data from annual quadrat maps, (ii) fit statistical models for survival, growth and recruitment, (iii) build population models based on the statistical vital rate functions and (iv) simulate the models to

**Table 1.** Summary of theoretical predictions relating indirect effects to niche differences. In the annual plant model, $N_{i,t}$ is the density of seeds of species $i$ at time $t$, $N_i^*$ is the steady-state density of species $i$, $\lambda$ is fecundity, and $\alpha_{ij}$ is the per capita effect of species $j$ on species $i$. For this example, we assume that the $\alpha$'s will not be influenced by climate, so the direct effect of climate change in this model is represented by an alteration in $\lambda$'s. The resulting changes in densities will then drive indirect effects. In the multispecies model, $\theta_i$ is a parameter of species $i$ that is assumed to be directly affected by climate change, which can be anything other than one of the competition coefficients $\alpha_{ij}$. For size-structured populations, $M_j(t)$ is a size-weighted measure of total population abundance of species $j$, and $T_i$ is a density-dependent operator describing how survival, growth, fecundity and competition change the species' abundance and size distribution from one year to the next. See Appendix A for a complete explanation of model assumptions and derivations of the predictions

| Model | Prediction | Interpretation |
|---|---|---|
| Two species model for annual plant competition (Harpole & Suding 2007; Levine & HilleRisLambers 2009):<br>$N_{1,t+1} = \frac{\lambda_1 N_{1,t}}{1 + \alpha_{11} N_{1,t} + \alpha_{12} N_{2,t}}$<br>$N_{2,t+1} = \frac{\lambda_2 N_{2,t}}{1 + \alpha_{22} N_{2,t} + \alpha_{21} N_{1,t}}$ | $\frac{\partial N_1^*}{\partial \lambda_2} = \frac{-\alpha_{12}}{\alpha_{11}\alpha_{22} - \alpha_{12}\alpha_{21}}$ | The effect of $\lambda_2$ on $N_1^*$ depends on the per capita effect of species 1 on species 2, scaled by the difference in the products of the intra- and interspecific terms. The more conspecifics limit themselves relative to heterospecifics, the weaker the potential strength of indirect effects of climate change. |
| General multispecies competition model for unstructured (a) or size-structured (b) populations:<br>(a) $N_i(t+1) = N_i(t) F_i\left(\sum_{j=1}^{n} \alpha_{ij} N_j, \theta_i\right)$<br>(b) $N_i(t+1) = T_i\left(\sum_{j=1}^{n} \alpha_{ij} M_j(t), \theta_i\right) N_i(t)$ | $\frac{\partial \bar{N}_i^*}{\partial \theta_j} \propto \alpha_{ij}/\alpha_{ii}$<br>$\frac{\partial \bar{N}_i^*}{\partial \alpha_{jr}} \propto \alpha_{ij}/\alpha_{ii}$ | The species with the strongest intraspecific competition ($\alpha_{ii}$) relative to interspecific competition ($\alpha_{ij}$) will be the least sensitive to indirect effects resulting from a direct effect on another species $j$, or to a change in the interaction between a pair of other species ($j$ and $r$). |



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estimate direct and indirect effects of climate change as well as niche differences. We briefly describe each of these steps below and provide full detail in Appendix S2. This analysis builds on Dalgleish *et al.* (2011), which focused on the role of climate covariates in single-species integral projection models (IPMs), and Adler, Ellner & Levine (2010), which took a multi-species approach but ignored climate drivers.

## EXTRACTING DEMOGRAPHIC DATA

The U.S. Sheep Experiment Station (USSES) is located 9.6 km north of Dubois, Idaho (44.2° N, 112.1° W), 1500 m above sea level. During the period of data collection (1926–57), mean annual precipitation was 270 mm and mean temperatures ranged from −8 °C (January) to 21 °C (July). The vegetation is dominated by the shrub, *Artemisia tripartita*, and the C3 perennial bunchgrasses *Pseudoroegneria spicata*, *Hesperostipa comata* and *Poa secunda*. These four species, the focus of our models, accounted for over 70% of basal cover (grasses) and 60% of canopy cover (shrubs and forbs).

Scientists at the USSES established 26 1-$m^2$ quadrats between 1926 and 1932. Eighteen quadrats were distributed among four ungrazed exclosures, and eight were distributed in two pastures grazed at medium intensity from spring (April) through fall (October). All quadrats were located on similar topography and soils. In most years until 1957, all individual plants in each quadrat were mapped using a pantograph (Blaisdell 1958). Digitized versions of the original maps are available online (Zachmann, Moffet & Adler 2010). Our models are based on data from 22 year-to-year transitions between 1926 and 1957. For the first two transitions, only four quadrats were observed, while at least 16 quadrats were observed for all subsequent transitions.

We extracted demographic data from the mapped plots by tracking individuals based on their spatial locations in the quadrats (Lauenroth & Adler 2008; Adler, Ellner & Levine 2010). Each mapped polygon with a buffered area that overlaps with genets present in previous years was classified as a survivor, inheriting the identity of the existing genet with the greatest overlap. Polygons with no overlap with existing genets were classified as new recruits. Our model represents individual genets as circles with two attributes, area (basal cover for grass, canopy cover for the shrub) and a spatial location (Adler, Ellner & Levine 2010).

## STATISTICAL MODELS OF VITAL RATES

We model the survival probability of each genet as a function of genet size, temporal variation among years, permanent spatial variation among quadrat locations and local neighbourhood crowding from both conspecific and heterospecific genets. Temporal variation among years is introduced through climate covariates as well as random year effects. The crowding experienced by the focal genet depends on the size of and distance to all neighbouring genets. We use a Gaussian function to describe how neighbour influence declines with distance (Appendix S2). The spatial scale over which individuals influence competitors varies among species, but we found that for each species we could use the same scale for growth and survival with little loss of fit (Appendix S2; Adler, Ellner & Levine 2010). Fitted regression coefficients determine how crowding by each species influences the focal individual's survival probability. We can present these regression coefficients as a matrix of interaction coefficients, with each *j,k* entry giving the effect of crowding by species *k* on the survival probability of a genet of species *j*. We fit these models in a generalized linear mixed-effects framework to accommodate random effects of year and quadrat group location (Appendix S2).

Our model for growth, the change in size of a genet from one year to the next, depends on the same factors as the survival model, with crowding effects described by the coefficients of a growth interaction matrix. We followed the same model selection and model fitting approach used for the survival models (Appendix S2).

To select the climate covariates for the survival and growth models, we began with the following variables: Annual precipitation in the year preceding the observed year-to-year transition (e.g. annual precipitation in year $t - 1$ for survival from year $t$ to $t + 1$), and fall–spring precipitation and mean spring temperature in both the first and second year of the transition. We also considered the interaction between fall–spring precipitation and spring temperature in each year. We chose these climate variables *a priori*, since the critical abiotic controls on plant growth in sagebrush steppe are soil moisture and temperatures during the short April–June growing season. We used a stepwise variable selection procedure based on Akaike's Information Criterion to select the best combination of these five initial climate covariates and two interactions. We allowed the climate covariates to influence both the intercept of the regression model and the effect of genet size on survival. We repeated this same procedure for the growth models. Note that this procedure requires large samples sizes, which we had, ranging from 1163 records for the *A. tripartita* growth regression to 7598 records for the *P. spicata* survival regression. Summaries of all the survival and growth models are included in Appendixes S3 and S4, respectively.

While the survival and growth regressions operate at the level of individual genets, we model recruitment at the quadrat scale because we cannot determine which recruits were produced by which parent genets. We assume that the number of recruits produced in each year in each quadrat follows a negative binomial distribution with a mean that depends on the cover of the 'parent' species, temporal variation among years due to fixed effects of climate covariates as well as random year effects, random effects of permanent spatial variation among groups of quadrats and both intra- and interspecific interactions dependent on each species' total cover in the quadrat (Appendix S2). We used the same five climate covariates as for survival and growth, but did not fit the precipitation × temperature interactions as they slowed convergence. We fit the recruitment parameters using a hierarchical Bayesian approach implemented in WinBUGS 1.4 (Lunn *et al.* 2000). Appendix S5 contains summary information for the recruitment model.

We conducted an additional analysis to provide some indication of the proportion of the interannual variation in vital rates explained by the climate covariates. We fit (i) a 'constant' model with no year effects or climate covariates (i.e. no temporal variation); (ii) a 'climate' model with climate covariates, but no random year effects, to explain temporal variation; and (iii) a 'full' model using climate covariates and random year effects to explain temporal variation. We calculated the portion of temporal variability explained by the climate covariates as

$$(X_{\text{climate}} - X_{\text{constant}})/(X_{\text{full}} - X_{\text{constant}})$$

where $X$ was the sum of the squared residuals, for the growth regressions, or the residual deviance, for the survival and recruitment regressions.

## MULTISPECIES POPULATION MODELS

The vital rate regressions are the building blocks of two multispecies population models (Appendix S2): an individual-based model (IBM)



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that we use to compare observed and simulated dynamics and an IPM that we use to simulate equilibrium abundances under historical and perturbed climate.

To reproduce the observed dynamics on the mapped quadrats, we initialized the IBM with the observed plant sizes and locations in the first mapped year for each quadrat, and then projected the modelled quadrat forward in chronological time, re-initializing the model after each gap in the observed data. At each time step in the simulation, we use the survival regression to determine whether each genet lives or dies, the growth regression to determine changes in size of surviving genets and the recruitment regression to calculate the production of new individuals, which are distributed randomly in space. We simulated each quadrat 100 times and averaged species' predicted cover and density in each year across the replicate simulations. We compared a simulation of the full model, which includes both random year effects and fixed effects of climate covariates on vital rates, with a second simulation in which we dropped the random year effects, using only climate covariates to drive interannual variation.

While the IBM works well for comparing the model to data, computing power limits the size of the simulated landscape and number of individuals. As a result, demographic stochasticity complicates efforts to describe long-term model behaviour by causing local extinctions. Therefore, we use a multispecies IPM (Easterling, Ellner & Dixon 2000; Ellner & Rees 2006, 2007) to better address questions about the asymptotic behaviour of large populations. The IPM is based on the same vital rate regressions used in the IBM (Adler, Ellner & Levine 2010). However, because the IPM is spatially implicit, we cannot calculate neighbourhood crowding for each individual. Instead, we use a mean field approximation that captures the essential features of the observed, non-random spatial pattern and results in equilibrium abundances that match the IBM projections (Appendix S2; Adler, Ellner & Levine 2010).

We used the IPM to simulate the community response to changes in climate covariates over long time-scales. We modelled temporal environmental variation using both climate covariates and the fitted random year effects. At each time step, we randomly selected climate covariates specific to one of the 22 observed years, and also selected one year of estimated survival, growth and recruitment random year effects (because the random year effects are uncorrelated with the climate covariates, we selected the climate covariate year independently of the random effects year). This approach, an analogue of the 'matrix-selection' technique used in stochastic simulations of structured population models, preserves observed covariances in climate covariates and demographic parameters within and among species. Based on little evidence of temporal structure in the time-varying demographic rates, the simulations ignore temporal autocorrelation. The simulations also ignore permanent spatial variation among quadrats, and thus represent dynamics on an idealized, 'average' site.

## DIRECT AND INDIRECT EFFECTS OF CLIMATE PERTURBATIONS

We used two perturbations of the IPM to partition the overall response of community composition into contributions from direct and indirect effects. First, we simulated the 'full effect' of climate change by making a 1% increase in the observed precipitation, temperature, or precipitation and temperature covariates, allowing all four species to respond simultaneously. Using this small perturbation to represent climate change keeps the model within the range of historical variation. Because our goal was to test our hypothesis about niche differences and indirect effects, we did not want to use a large perturbation that would raise questions about extrapolating a model based on observational data. Given predictions that climate change will increase climate variability (Christensen *et al*. 2007), and our own work showing that some of these species respond to temporal variability (Adler, Ellner & Levine 2010), we conducted a fourth perturbation involving climate variances. Without altering the precipitation and temperature means, we increased the standard deviation of the observed climate time series by 10% (changes in abundance caused by a 1% increase in standard deviation were difficult to detect). This first simulation includes both the direct and indirect effects of climate change, thus the resulting changes from species' baseline equilibrium cover represent the full effect of climate change.

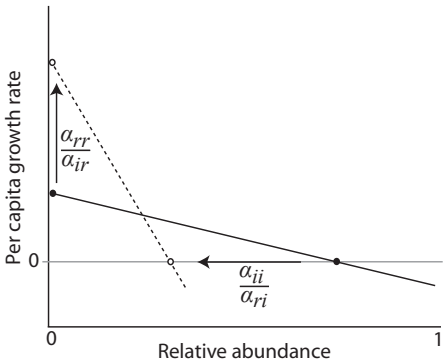



**Fig. 1.** Negative frequency dependence as a measure of niche differences. For a focal species *i*, the slope of the line relating its (log) per capita growth rate to its frequency in the community is a measure of the difference between the strength of intra- and interspecific competition and therefore a measure of niche differentiation for that species (also see Appendix A). The effect of the competitor species *j* on itself, $\alpha_{jj}$, and its effect on the focal species, $\alpha_{ij}$, influence both the low-density growth rate (the y-intercept) and the equilibrium abundance (the x-intercept) of the focal species. In contrast, the effect of the focal species on itself, $\alpha_{ii}$, and its effect on the competitor species, $\alpha_{jj}$, only influence relative abundance. Therefore, an increase in the ratio $\alpha_{ii}/\alpha_{ji}$ must steepen the slope of the relationship and reflects an increase in self-limitation relative to limitation of heterospecifics. As self-limitation grows stronger (e.g. the shift from the solid black to the dashed line), the focal species becomes less sensitive to interspecific competition. Therefore, species with steep (negative) slopes in a plot of per capita growth rate versus relative abundance should experience weak indirect effects of climate change.

Second, we simulated the 'direct effect' of climate change by focusing on one species at a time. In this simulation, the vital rates of the focal species were determined by the perturbed climate variable(s) and by the corresponding perturbation to intraspecific crowding. However, the interspecific crowding that the focal species experienced in each simulated year were taken from a parallel simulation of the community using unperturbed climate. Thus, all interspecific effects involving the focal species are held at their unperturbed levels, preserving the temporal variability in heterospecifics' cover and size distributions characteristic of the sequence of years being simulated (failing to properly account for this interannual variability in competition had a large effect on simulated abundances). This simulation represents the direct effect of the climate perturbation because the intrinsic performance of the focal species is altered but interactions with other species are not. Therefore, we defined the 'indirect effect' of climate on each species as the difference in average cover between the 'full effect' and 'direct effect' simulations. For all simulations, both 'full' and 'direct', we used the same sequence of 2250 randomly generated climate years and random effect years, removing the first 250 steps before averaging abundances.



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### QUANTIFYING NICHE DIFFERENCES

In contrast to traditional approaches that focus on specific coexistence mechanisms or niche axes (e.g. Janzen 1970; Connell 1971; Whittaker, Levin & Root 1973; Tilman 1982; Silvertown *et al.* 1999; McKane *et al.* 2002), our goal is to quantify the overall, demographic effect of all niche differences. For any given focal species, the slope of the line relating its per capita growth rate to its frequency in the community is a measure of the difference between the strength of intra- and interspecific competition (Fig. 1), and therefore a measure of niche differentiation for that species (Adler, HilleRisLambers & Levine 2007; Levine, Adler & HilleRisLambers 2008). This negative frequency dependence is the demographic outcome of all stabilizing niche differences, whether they are based on differential use of resources, spatial or temporal heterogeneity, or species-specific pathogens (Chesson 2000).

We relied on two different simulations of the IPM to describe each species' pattern of negative frequency dependence (Appendix S6 explains why we relied on simulations, rather than calculating frequency dependence directly from field data). First, we simulated equilibrium abundances by initializing all species at low abundance and running the model for thousands of time steps (drawing randomly from the 22 sets of climate covariates and random year effects at each time step). Second, we simulated each species' low-density growth rate by initializing all species at low abundance, but while we allowed the 'resident' species to reach equilibrium, after each time step we returned the focal species to such low abundance that it had no effect on itself or any other species (Adler, Ellner & Levine 2010). We repeated this process for each species. The slope of the line connecting the low-density growth rates with the equilibrium frequencies provides an estimate of negative frequency dependence. The prediction we test is that species with weak negative frequency dependence should be most sensitive to the indirect effects of climate perturbations.

## Results

### CLIMATE EFFECTS ON DEMOGRAPHY AND DYNAMICS

Before using our multispecies population model to estimate the indirect effects of climate perturbations, we needed to verify that the climate covariates in our model are statistically and ecologically important (Appendix S7 shows parameter values for all species and vital rate regressions). We assessed their statistical importance by comparing the residual deviance or sums of squares of vital rate regressions with no temporal variation, only climate covariates, or climate covariates and random year effects together (accounting for the maximum temporal variability). This comparison shows the portion of the total interannual variability in vital rates explained by the fitted climate covariates. For the survival regressions, the climate covariates reduced residual deviance by 28–75% of the maximum possible reduction in deviance (Table 2). For growth, the climate covariates were less effective, reducing the sums of squared residuals by 22–56% of the maximum possible. In the recruitment model, which we fit simultaneously for all four species, climate covariates reduced the deviance by 27%. The large residual variation still present in the full models with random year effects reflects variation at the quadrat and individual plant level.

We assessed the ecological importance of the climate covariates in two ways. First, we used the IBM to simulate historical dynamics on each quadrat (as described in Adler, Ellner & Levine 2010). Simulations that included both random year effects and fixed effects of climate covariates successfully reproduced observed dynamics of both cover and density (Fig. 2). When we repeated this simulation using only the climate covariates to model temporal variation (with the random year effects turned off), the simulations were still plausible for cover but often failed to capture boom years in recruitment which show up as spikes in density (Fig. 2). This result is consistent with the relatively weak statistical effects of climate covariates in the recruitment regression. Second, we used the IPM to compare the interannual variability in species' cover in simulations with observed climate variation and simulations with climate covariates held constant at their mean values. The constant climate simulation reduced the standard deviation of

**Table 2.** The proportion of interannual variability in vital rates explained by the climate covariates. The 'Constant model' does not allow for any variability in vital rates, the 'Climate model' explains temporal variability using climate covariates only, and the 'Full model' explains temporal variability using both climate covariates and random year effects

| | *N* | Constant model | Climate model | Full model | Contribution of climate covariates* |
|---|---|---|---|---|---|
| Growth | Sum of squared residuals | | | | |
| *Artemisia tripartita* | 1163 | 1171.1 | 1105.9 | 874.5 | 0.22 |
| *Hesperostipa comata* | 2639 | 1724.98 | 1555.2 | 1419.6 | 0.56 |
| *Poa secunda* | 3406 | 2912.5 | 2819.4 | 2672.5 | 0.39 |
| *Pseudoroegneria spicata* | 4884 | 3227.9 | 3107.9 | 2855.9 | 0.32 |
| Survival | Residual deviance | | | | |
| *A. tripartita* | 2058 | 1293.1 | 1200.7 | 1169.9 | 0.75 |
| *H. comata* | 3818 | 2639.6 | 2511.52 | 2409.6 | 0.56 |
| *Po. secunda* | 4992 | 4014.4 | 3911.4 | 3645.1 | 0.28 |
| *Ps. spicata* | 7598 | 4366 | 4080.4 | 3817.1 | 0.52 |
| Recruitment | Residual deviance | | | | |
| All species | 1664 | 4017.9 | 3842.4 | 3229.7 | 0.27 |

*(Climate model − Constant model)/(Full model − Constant model).



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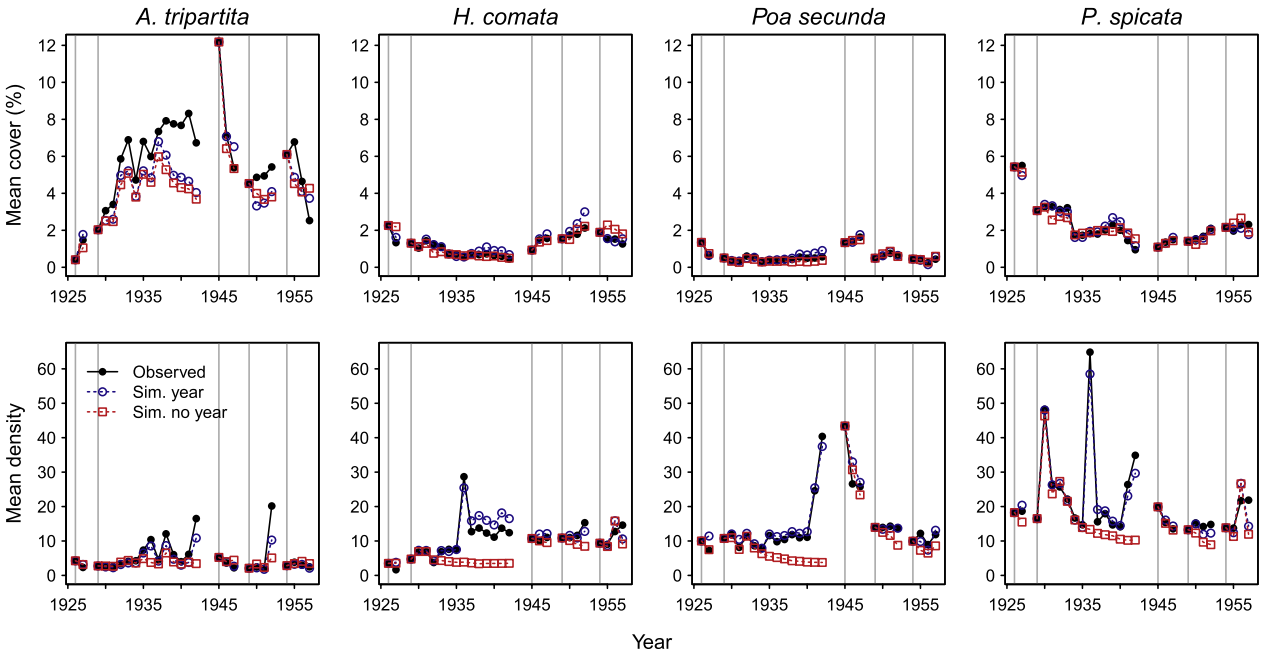



**Fig. 2.** Observed (solid symbols) and predicted mean cover (top row) and density (bottom row) of the four modelled species (columns; note differences in y-axis ranges). Predictions come from simulations of an individual-based model using either climate covariates and random year effects to drive interannual variation in performance (open circles) or only climate covariates (open squares). For each of the 26 mapped permanent quadrats, the model was initialized with the observed genet sizes and spatial coordinates at dates indicated by the vertical grey lines and then projected forward until the next data gap. Shown are means across 100 simulation runs for each quadrat. Error bars show the 10th and 90th percentiles of the simulation runs.

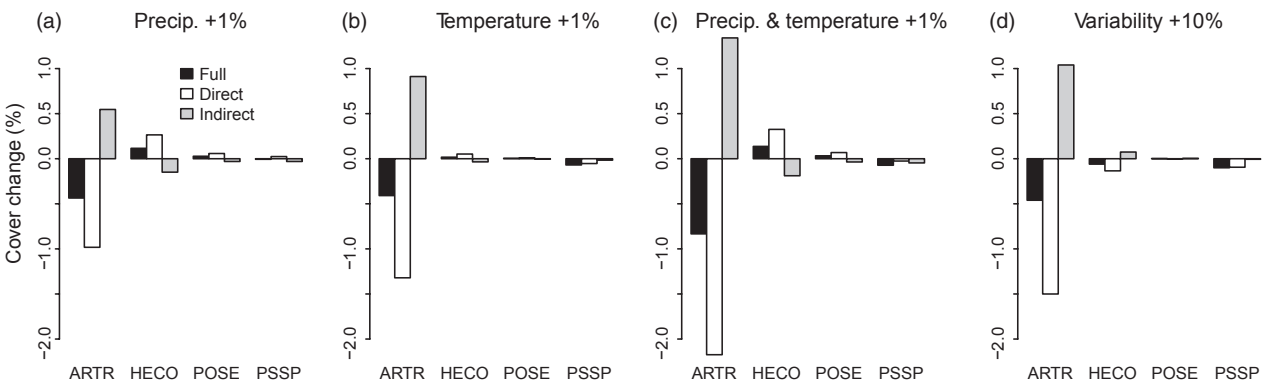



**Fig. 3.** Projected absolute changes in species' average cover caused by 1% increases in historical, observed precipitation (a), temperature (b) or precipitation and temperature (c) and a 10% increase in the standard deviation of precipitation and temperature (d).

species' cover by only 1% for *Ps. spicata* but 22–30% for the remaining three species. Decreases in the coefficient of variation were larger, from 29 to 57%, reflecting increases in species' mean abundances. Both of these analyses indicate that the fitted climate covariates have substantial effects on community dynamics.

## DIRECT AND INDIRECT EFFECTS OF CLIMATE PERTURBATIONS

Simulated equilibrium cover was sensitive to climate. The proportional changes in cover resulting from the different climate perturbations ranged from 13 to 27% for *A. tripartita*, 1–7% for *H. comata* and *Ps. spicata*, and 1–4% for *Po. secunda* (Fig. 3).

A 1% increase in the means of the precipitation covariates caused a decrease in the equilibrium cover of *A. tripartita*, a small increase in *H. comata* cover and very little change in the cover of *Po. secunda* and *Ps. spicata* (Fig. 3a). For the three grasses, the change in cover represented a positive direct effect of increased precipitation offset by strong, negative indirect effects. In contrast, the overall decrease in *A. tripartita* reflected a strong negative direct effect, offset by positive indirect effects.



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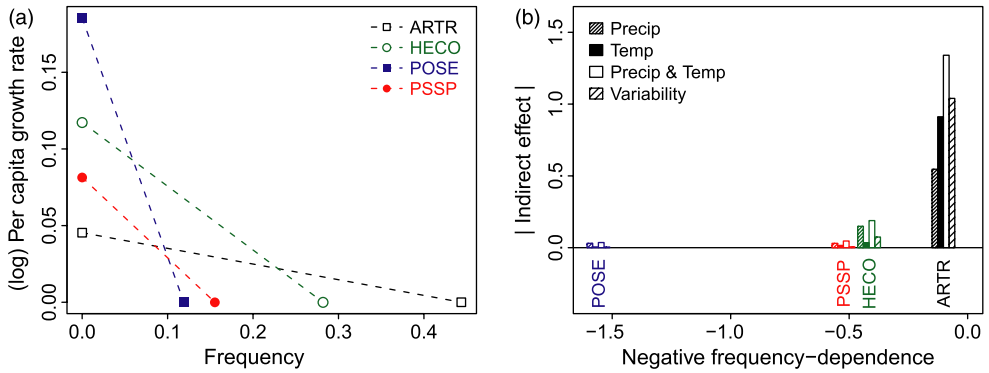



**Fig. 4.** (a) Negative frequency dependence in per capita growth for the four study species and (b) the relationship between the slope of negative frequency dependence and the absolute value of indirect effects of precipitation ('Precip'), temperature ('Temp') and variability perturbations on each species. Although the difference is hard to see, *Ps. spicata* experienced a stronger maximum indirect effect of the precipitation plus temperature perturbation than did *Poa secunda* (0.047 vs. 0.036).

A 1% increase in spring temperatures led to similar qualitative patterns, though compared with the precipitation perturbation the size of the effects were larger for *A. tripartita* and smaller for the grasses (Fig. 3b). The outcome of perturbing precipitation and temperature together appeared to be the sum of the single variable perturbations (Fig. 3c).

A 10% increase in the standard deviations of all climate covariates had a negative direct effect on all four species, with *A. tripartita* showing the largest change and *Po. secunda* the smallest. In this case, the indirect effects were positive for all species except *Ps. spicata,* with changes in community composition buffering these species against the negative direct effects of variability.

### NICHE DIFFERENCES AND INDIRECT EFFECTS

All four species showed negative frequency-dependent patterns of population growth, although the strength of the frequency dependence varied among species (Fig. 4a). Every species had a positive growth rate when rare, indicating stable coexistence. *Poa secunda* had the highest invasion growth rate but the lowest equilibrium abundance, leading to very strong negative frequency dependence. In contrast, *A. tripartita* had the lowest invasion growth rate but the highest equilibrium abundance, leading to relatively weak frequency dependence. *H. comata* and *Ps. spicata* showed intermediate patterns.

To test our hypothesis that weak niche differences lead to strong indirect effects, we plotted the absolute value of the simulated indirect effects (from Fig. 3) against our proxy for stabilizing niche differences, the slope of negative frequency dependence for each species (from Fig. 4a). Consistent with our hypothesis, *A. tripartita*, the species with the weakest negative frequency dependence, experienced the strongest indirect effects, while *Po. secunda*, the species with the strongest negative frequency dependence, was the least sensitive to indirect effects (Fig. 4b). Although the direction of the indirect effects depended on the idiosyncracies of each species' direct response to the climate covariates as well as the direction and strength of pairwise species interactions, the potential strength of indirect effects decreased rapidly with stronger niche differences.

## Discussion

### NICHE DIFFERENCES AND THE INDIRECT EFFECTS OF CLIMATE CHANGE

Our analysis of the empirical, multispecies population model supported our hypothesis: Species with dynamics strongly stabilized by niche differences experienced the weakest indirect effects of climate, while the species most weakly stabilized by niche differences was most sensitive to indirect effects. In theoretical models, the relationship between niche differences and indirect effects is easy to demonstrate (Table 1 and Appendix S1). Although our empirical analysis included only four species, the emergence of the same relationship in a model fit with observational data shows that the theory may be useful for understanding natural communities. In fact, the strength of indirect effects appears to increase nonlinearly with decreases in niche differences (Fig. 4b), which is exactly what the theory predicts (Appendix S1, eqn 2).

While our hypothesis explains the relative magnitudes of indirect effects within a community, it does not explain their absolute magnitudes or their directions. Indeed, we found strong contingencies in the indirect effects of climate perturbations projected by our model: Increased precipitation and temperature had negative direct effects on the shrub, *A. tripartita*, but positive direct effects on the grasses. Normally, this scenario would lead to strong indirect effects on the shrub, so the positive indirect effect of precipitation on *A. tripartita* surprised us (Fig. 3). The explanation is that *A. tripartita* has a facilitative effect on grass recruitment, so reductions in its cover reduce the cover of the grasses, decreasing competitive pressure on *A. tripartita*. Our simulation of the direct effect of precipitation on *A. tripartita* held grasses at their higher, baseline level, exacerbating the negative direct effect of precipitation on *A. tripartita*. Such complex patterns will arise from differences among species, and even among vital rates, in the strength and direction of the climate effects and the interspecific interaction coefficients. An important result of our study is that the tight relationship between niche differences and strength of indirect effects emerged despite these idiosyncrasies.



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A focus on niche differences may not solve the forecasting challenge but can greatly simplify it by helping us to determine when we can safely ignore interspecific interactions. For species strongly stabilized by niche differences, such as the bunchgrasses in our study system, multispecies models may not be necessary. Instead, single-species models, which treat interactions implicitly, should be adequate. Similarly, information about niche differences could help managers determine when mulitspecies approaches are necessary to meet conservation objectives (Nicholson & Possingham 2006).

For the three grass species in our model, niche differences were strong and the indirect effects of climate perturbations were very weak. Unfortunately, we do not know whether the exceedingly stable coexistence of the bunchgrasses in this sagebrush steppe community (Adler, Ellner & Levine 2010), and their insensitivity to indirect effects, represents a general pattern or an exceptional case. Niche differences in our system appear quite strong compared with results from short-term, experimental manipulations of annual plants (Levine & HilleRisLambers 2009). Further comparisons will not be possible until the stabilizing effect of niche differences has been fully quantified in other natural communities. Determining the importance of niche differences relative to neutral or equalizing forces (Adler, HilleRisLambers & Levine 2007) is not just a theoretical problem, but has implications for ecological forecasting. The irony is that in communities with more neutral interactions (where intra- and interspecific effects are of similar magnitude) species interactions will be more important in mediating the impact of climate change.

### LIMITATIONS AND CHALLENGES

We have argued that information about niche differences is useful in forecasting the ecological impacts of climate change. But as the previous discussion shows, in most natural communities we have essentially zero information about the stabilizing strength of niche differences (Siepielski & McPeek 2010). Long-term, community-wide demographic data sets such as the one we used to build our multispecies model are extremely rare. To apply our generalization about niche differences to a broad range of communities, we need a way to predict niche differences using easily collected data.

Both experimental and observational approaches are possible. For communities of organisms with simple life cycles, experiments using synthetic communities have described patterns of frequency dependence, our proxy for niches differences (Harpole & Suding 2007; Levine & HilleRisLambers 2009). An observational approach could rely on relationships between functional traits and pairwise interaction coefficients (Freckleton & Watkinson 2001; Berlow *et al*. 2009). While independent estimates of interaction effects are necessary to fit these relationships, general patterns that emerge could be broadly applied. Another alternative might focus on intra- and interspecific differences in plant–soil feedbacks, which appear to be powerful sources of density dependence (Klironomos 2002; Comita *et al*. 2010; Mangan *et al*. 2010) and may be correlated with simple traits (Kulmatiski *et al*. 2008).

A second limitation of our approach is that it assumes that indirect effects of climate change are mediated by changes in species abundances, rather than changes in the strength or direction of per capita interaction effects. While some experimental studies have shown that climate change could alter per capita interaction effects among plant species (Klanderud 2005; Saccone *et al*. 2009), others have failed to show that climate variability influences interaction coefficients (Greenlee & Callaway 1996; Tielborger & Kadmon 2000; Adler, Leiker & Levine 2009; Adler, Ellner & Levine 2010; Levine, McEachern & Cowan 2010). Even if climate alters per capita interaction effects, our theoretical work indicates that niche differences will still determine species' sensitivity to such changes: Species with strong niche differences will be less sensitive to a change in another species' competitive effect (Appendix S1).

Finally, our phenomenological approach for linking demography with climate provides little information about physiological mechanisms. For example, our model predicts that *A. tripartita* responds negatively to precipitation. *Artemisia* is probably not responding directly to precipitation or soil moisture, but rather to another variable correlated with high precipitation. For example, snow mould fungi can decrease *Artemisia* growth and survival during wet winters with deep snowpacks (Hess, Nelson & Sturges 1985; Allen, Allen & West 1987). Because such hidden correlations are implicit in our model projections, experimental validation will be essential before using these models in a true forecasting context.

## Conclusion

Robust generalizations about the role of species interactions in mediating climate change impacts do exist, hiding in plain sight in the theoretical literature. However, applying these generalizations to ecological forecasting remains a challenge. The inverse relationship between niche differences and the strength of indirect effects that we have demonstrated will not be of practical use until we develop ways of predicting niche differences from simple traits. This long-standing goal of basic research should also be a priority for conservation and management.

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## Supporting Information

Additional Supporting Information may be found in the online version of this article:

**Appendix S1.** Mathematical support.

**Appendix S2.** Detailed methodology.



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**Appendix S3.** Statistical results for the survival models for each of the four species.

**Appendix S4.** Statistical results for the growth models for each of the four species.

**Appendix S5.** Statistical results for the recruitment model.

**Appendix S6.** Why we estimate negative frequency-dependence with model simulations rather than field data.

**Appendix S7.** Parameter estimates for the survival, growth and recruitment models.

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